RegioManip: Region-Level Affordances for Zero-Shot Wrench-Bounded Tool Manipulation

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ABSTRACT

Contact-rich tool manipulation remains a last-mile challenge for embodied artificial intelligence and industrial robotics, because many real tasks require sustained physical interaction rather than simple pose reaching. Recent vision-language model (VLM) pipelines improve open-vocabulary grounding and instruction following, but their keypoint- or waypoint-centric outputs often commit to a single prediction and do not explicitly encode contact-frame stability, wrench limits, or perceptual uncertainty. We present RegioManip, an AI-driven zero-shot framework that converts grounded tool regions into wrench-bounded action constraints for contact-rich manipulation. Given a single RGB-D observation and a natural-language instruction, RegioManip grounds each task-relevant region into grasp-type, contact-geometry, force-mode, and contact-face attributes, cross-checks them against geometry-only estimates, and prunes infeasible combinations through a physics-consistency lattice. The resulting ranked affordance hypotheses are lifted into SE(3) contact-frame hypotheses and executed with force and torque budgets derived from the grasp wrench space of the resolved contact. Confidence-adaptive strictness and hierarchical repair enable robust execution under perceptual uncertainty. On a JAKA Zu3 across six industrial tool operations, RegioManip achieves 80.8% average open-world success, compared with 65.0% for GeoManip and 57.5% for MOKA, and reaches 90.8% success with a FoundationPose prior versus 86.7% for ToolManip.

Nomenclature

English symbols

I, D, K	RGB image, depth map, camera intrinsics
ℓ	Natural-language instruction
S, τ_i	Step sequence, step i
a_p	RegionAffordanceTuple for region p
r_p, g_p, c_p, f_p	Role, grasp type, contact geometry, force mode
b_p, s_p	Contact-face selector, semantic label
u_p^k	Per-attribute uncertainty, attribute k
G, C^*	Grip point, contact point
w_g	Metric grasp width
$\hat{a}$, $\hat{n}$, $\hat{t}$	Tool axis, contact-surface normal, tangent
$\mathcal{H}^+$, h^*	Contact-frame hypothesis set; selected hypothesis
c_i	Composite confidence score for hypothesis i
$\mathcal{W}_i$	GWS-derived admissible wrench set
$w_{\max}$	Confidence-scheduled admissible wrench bound
$w_{\min}$, w_{nom}	Conservative and nominal endpoints of admissible wrench bound
T	Robot pose in SE(3)

Greek symbols

ρ_p	Aggregate per-region confidence
η_k^{vlm}, $\bar{\eta}_{\text{vlm}}$	Per-attribute / mean VLM confidence
Ψ, ψ_k	Geometry–VLM agreement score
Λ	Physics-consistency lattice over (g, c, f) triples
Δ^+	Affordance simplex after geometry–VLM fusion
λ	Geometric strictness weight
λ_{nom}, λ_{safe}	Nominal and safe (tightened) endpoints of geometric strictness weight
Π_i	Action constraint program for step i
ϕ_i	Step success predicate
μ	Friction coefficient
ε_d, ε_Λ	Depth-consistency threshold, lattice weight

Abbreviations

VLM	Vision-language model
GWS	Grasp wrench space
SAM 2	Segment Anything Model 2
EMA	Exponential moving average
SLERP	Spherical linear interpolation
JSD	Jensen–Shannon divergence
F/T	Force/torque
PCA	Principal component analysis
SVD	Singular value decomposition
RGB-D	Red-green-blue + depth
SE(3)	Special Euclidean group in 3D

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1. Introduction

Recent advances in foundation models and vision-language models (VLMs) have significantly improved the ability of embodied artificial intelligence systems to interpret open-world scenes, understand natural-language instructions, and infer manipulation strategies without task-specific training. However, converting high-level multimodal reasoning into reliable physical interaction remains difficult when task success depends on accurate tool alignment, stable contact, and force-regulated execution rather than simple pose reaching [1–6].

Robotic tool manipulation remains a last-mile bottleneck for embodied intelligence and industrial automation, because many practical tasks are defined by sustained surface contact rather than goal reaching alone. Operations such as scraping, pressing, wiping, hammering, and screw loosening succeed only when the tool maintains a physically consistent interaction frame and respects force and torque limits throughout contact, not merely when it reaches a desired pose [1–3]. In this regime, millimeter-scale position

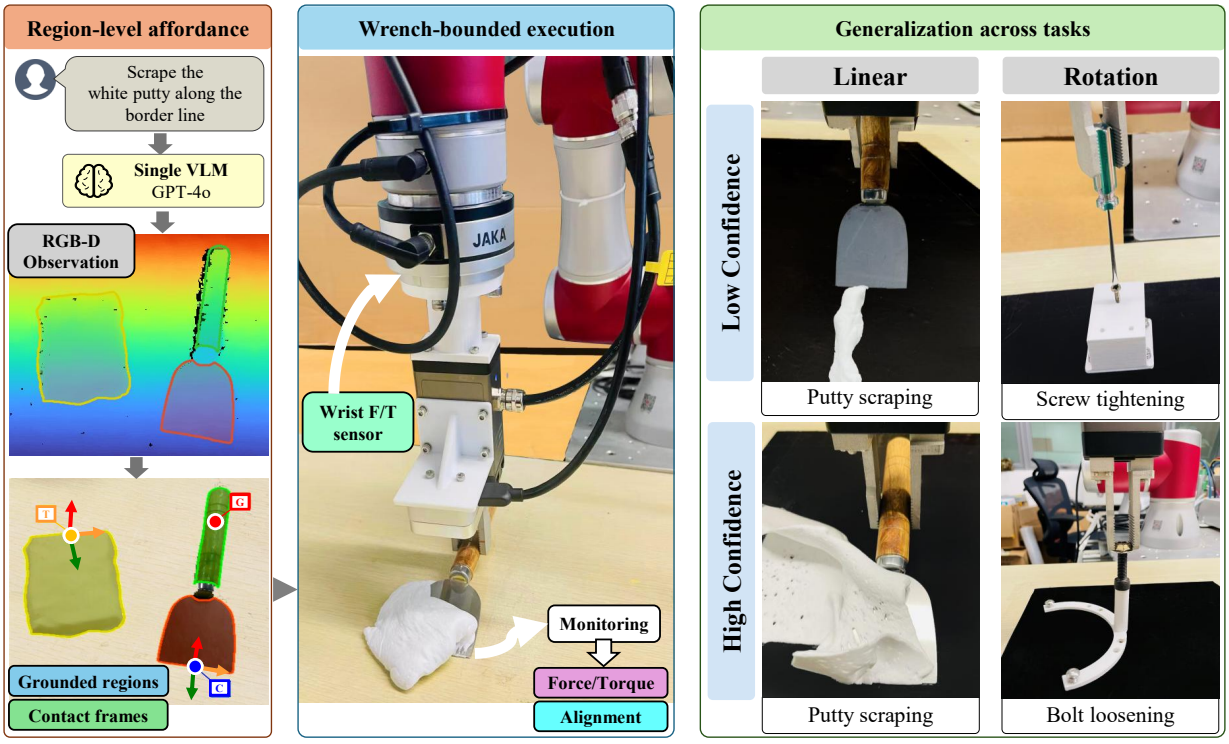


Figure 1: RegioManip overview: region-level affordance grounding converts tool regions into wrench-bounded action constraints, illustrated on linear- and rotational-contact tasks.

bias or small orientation drift can flip a contact normal, induce slip, jam motion, or trigger a force spike, so interaction geometry and wrench regulation, rather than goal reaching alone, determine success [4–6].

Existing manipulation stacks address contact through explicit modeling, hand-engineered task structure, and interaction controllers, but they generalize poorly across unseen tools, fixtures, and open-world scenes, especially when the relevant geometry is partially observed, uncertain, or changes during execution [4, 7]. Learning-based methods broaden task coverage, yet robust contact-rich tool execution remains difficult to acquire from data because diverse contact trajectories are costly to collect, safety-critical failures are rarely observed, and explicit wrench limits are hard to enforce online when contact states, friction regimes, and lever arms vary during execution [8–11]. These difficulties motivate AI-driven methods that can ground interaction structure directly from open-world observation and carry it into execution as an explicit, enforceable physical constraint.

Recent VLM-based robotic manipulation pipelines improve open-vocabulary grounding and instruction following, converting visual evidence and language instructions into actionable cues such as keypoints, waypoints, geometric constraints, or manipulation primitives [7, 12, 13]. For tool use, however, these intermediate representations remain under-specified for sustained contact: sparse keypoints and waypoints do not capture the region-level interaction structure needed to define contact pose, surface normal, and force direction, while constraint satisfaction alone cannot guarantee stable interaction when contact-frame regulation is absent [6, 8, 9]. Most zero-shot pipelines also commit to a single grounding prediction and pass it forward as a point estimate, discarding perceptual uncertainty and leaving no principled basis for recovery when that prediction is wrong [14, 15]. Moreover, the robustness of perceptual grounding under viewpoint variation, clutter, and novel tool instances remains underexplored.

To address these gaps, we propose RegioManip, an AI-driven zero-shot framework that converts grounded tool regions into wrench-bounded action constraints for contact-rich manipulation. Fig. 1 summarizes the overall workflow and the class of contact-rich tool operations targeted in this work. Given a single RGB-D observation and a natural-language instruction, RegioManip grounds each task-relevant region into a structured affordance representation including functional role, grasp type, contact geometry, dominant force mode, and contact-face selector. Contact primitives are then extracted from the grounded region geometry and lifted into ranked $\mathrm{SE}(3)$ contact-frame hypotheses that preserve orientation and lever-arm structure. The predicted attributes are checked against geometry-only estimates and filtered through a physics-consistency lattice, yielding a ranked set of mechanically feasible affordance hypotheses instead of a single commitment. During execution, RegioManip enforces force and torque budgets derived from the grasp wrench space of the resolved contact, with strictness and repair frequency modulated by hypothesis confidence. Overall, this design makes caution an explicit and interpretable part of the execution

interface, enabling mechanically informed manipulation beyond sparse keypoint- or waypoint-based grounding and predefined motion/force primitive templates.

We evaluate RegioManip on a JAKA Zu3 across six industrially motivated tool operations spanning linear and rotational motions and constant-force and threshold-force regimes. The evaluation further examines robust perceptual grounding under viewpoint changes, clutter, and novel tool instances. Two regimes make the pose-prior assumption explicit: an open-world regime with no pose prior or object model, compared against MOKA [12] and GeoManip [7], and a reference-assisted regime with a FoundationPose [16] prior, compared against ToolManip [4]. Beyond success rate, we report geometric drift and peak wrench utilization, showing that contact-frame constraints reduce drift and preserve larger wrench margins rather than compensating for misalignment through excessive force.

The main contributions of this work are as follows:

- **Reliability-aware region-level affordance grounding.** A single VLM call grounds each task-relevant region into grasp-type, contact-geometry, force-mode, and contact-face attributes. These predictions are cross-checked with geometry-only estimates and pruned by a physics-consistency lattice, yielding ranked affordance hypotheses rather than a single keypoint or waypoint. Contact primitives are then extracted from the grounded region geometry to expose interaction structure and support recovery under perceptual uncertainty.
- **Region-to-SE(3) contact-frame lifting.** We back-project the extracted contact primitives into metric 3D structure and fuse them across frames on their natural manifolds, producing ranked contact-frame hypotheses that preserve orientation and lever-arm structure. The result is a controller-consumable interface aligned with contact stability, rather than a sparse waypoint sequence.
- **Wrench-bounded, confidence-adaptive execution.** The admissible force and torque budget is derived from the grasp wrench space of the resolved contact, and hypothesis confidence determines how strictly the wrench budget and contact-frame constraints are enforced. Residual- and wrench-triggered hierarchical repair recovers when alignment stalls or the measured wrench approaches its bound, tying grounded interaction structure directly to enforceable wrench margins during sustained contact.

2. Related Work

We review related work on the perception-to-execution interface for open-world tool manipulation, covering foundation-model robotic pipelines, intermediate interaction representations, and safe contact-rich manipulation. RegioManip differs by linking region-level affordance grounding, SE(3) contact-frame hypotheses, and GWS-derived wrench-bounded execution under confidence-adaptive repair.

2.1. Foundation Models for Robotic Manipulation

Foundation models have substantially advanced robotic manipulation by improving semantic understanding, instruction following, and open-world generalization. Vision-language-action models such as RT-1 and RT-2 demonstrate that large-scale robot data and web-scale visual-language knowledge can be transferred into executable robot actions [17, 18]. PaLM-E and Open X-Embodiment further improve cross-task and cross-platform generalization by combining multimodal pretraining with large multi-robot datasets [19, 20], while generalist robot policies such as Octo aim to make these capabilities reusable across embodiments and tasks [21]. Diffusion-based policies improve multimodal trajectory generation in high-dimensional action spaces [22], and representation and tokenization choices continue to improve efficiency and scalability [23]. These methods greatly expand task coverage, but contact-rich tool manipulation remains difficult because the dominant failure modes are often physical rather than semantic, including contact-frame drift, slip, jamming, and excessive wrench generation [8–10].

A complementary line of work builds agentic and modular robotic systems that explicitly manage planning, execution, monitoring, and recovery. Agentic frameworks emphasize plan-execute-recover loops for scalable manipulation [24, 25], and UniManip couples high-level agentic reasoning with a low-level scene representation to diagnose and recover from execution failures across embodiments [26]. Local-policy decomposition improves long-horizon robustness by composing reusable closed-loop skills [27], while language-model programs provide a code-based interface for feedback-driven robot behavior [28]. Other systems stabilize decision-making through task-relevant memory [29], incorporate perception and language feedback into replanning [30, 31], or use VLM reasoning to constrain task-and-motion planning under uncertainty [32–34].

These pipelines improve robustness to partial observability, changing scenes, and long-horizon task structure. However, for tool use, the execution interface is often evaluated mainly through symbolic completion, pose reaching, or geometric feasibility. Stable tool interaction requires more: the robot must maintain a physically meaningful contact frame and explicit wrench margins throughout sustained contact. RegioManip therefore uses foundation-model grounding not only to infer what to do, but also to construct contact-frame hypotheses and enforce wrench-bounded execution during physical interaction.

2.2. Interaction Representations for Tool Use

The intermediate representation passed from perception to control is critical for open-world tool manipulation. Keypoint- and waypoint-based methods ground tasks into sparse spatial cues. MOKA uses mark-based visual prompting to predict actionable keypoints and waypoints that connect VLM outputs to robot motion [12], while

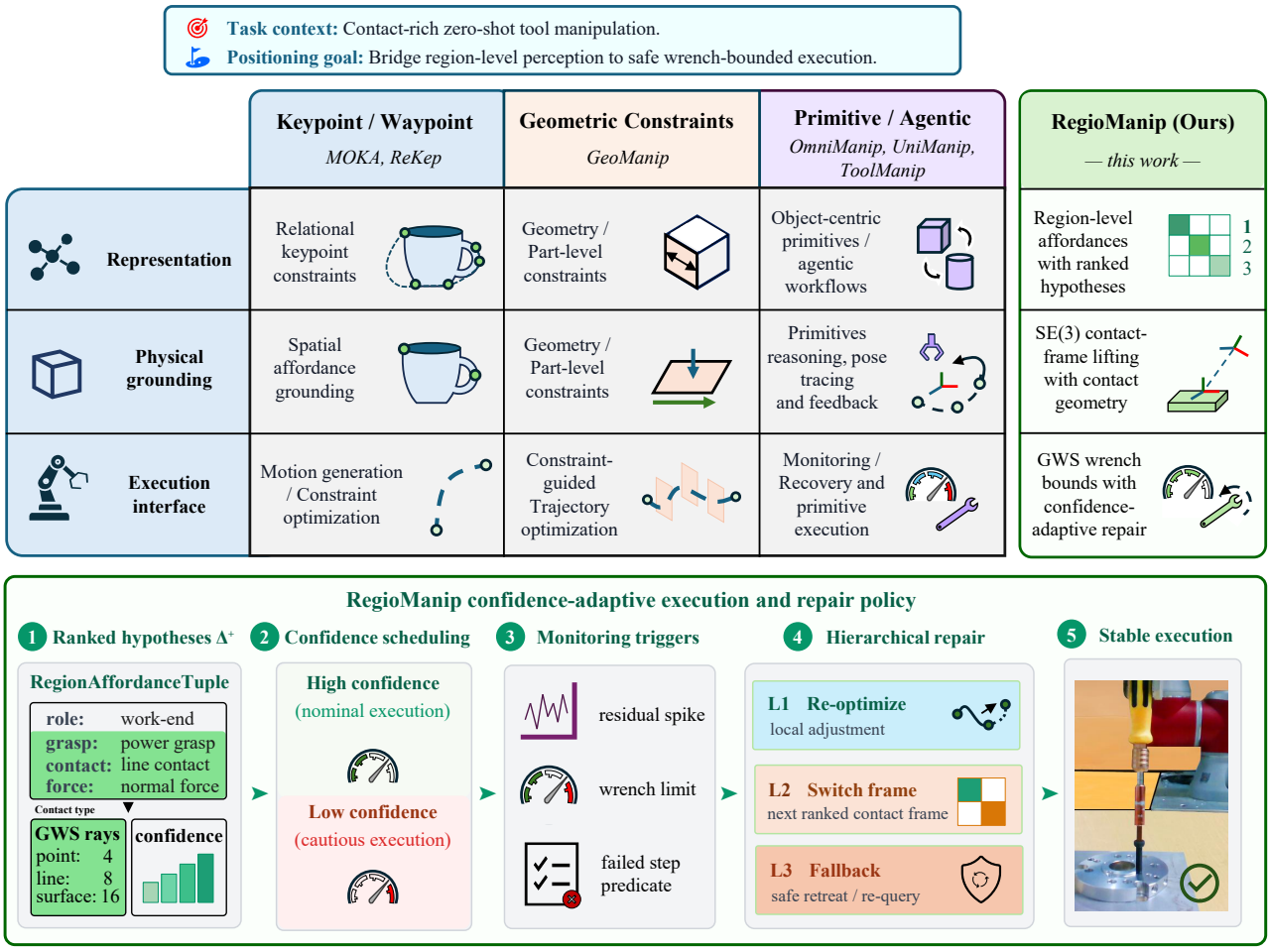


Figure 2: Positioning of RegioManip relative to representative prior methods. Keypoint and geometric-constraint methods typically rely on sparse or single-shot interaction representations, while primitive and agentic methods add feedback and recovery mechanisms. RegioManip combines region-level affordance grounding, ranked SE(3) contact-frame hypotheses, GWS-derived wrench bounds, and confidence-adaptive repair.

ReKep represents manipulation tasks as relational keypoint constraints optimized in a closed-loop manner [13]. These representations are compact and interpretable, but sparse points do not fully specify the contact pose, surface normal, force direction, or lever-arm structure required for sustained tool contact.

A second group of methods moves toward solver-friendly constraint interfaces. CoPa uses part-conditioned spatial constraints for task-aware manipulation [35], GeoManip formulates geometric constraints as general interfaces for trajectory optimization [7], OmniManip introduces object-centric interaction primitives as spatial constraints [36], and T-Rex studies task-adaptive representation selection among points, vectors, and poses [37]. ToolFlowNet provides a complementary tool-centric perspective by predicting tool flow from point clouds, showing that tool use benefits from interaction-centric geometry beyond end-effector poses [6]. These methods improve the expressiveness of robot manipulation interfaces, but constraint- or primitive-level satisfaction alone does not ensure stable interaction when the contact frame and admissible wrench are not explicitly regulated [6, 8, 9].

Another limitation of many existing representations is that they commit to a single grounding prediction and pass it forward as a point estimate. As a result, perceptual uncertainty is often discarded before execution. Reliability-aware tool use instead requires confidence-conditioned decisions under uncertainty [15, 38], while semantic monitoring shows that VLM-level checks can complement low-level feedback after execution failures [39]. RegioManip addresses this limitation by grounding tool regions into structured affordance attributes, cross-checking VLM predictions against geometry-only estimates, and forwarding a ranked distribution of mechanically feasible hypotheses rather than a single prediction.

At the perception level, open-vocabulary grounding and region understanding continue to improve. Open-world training paradigms expand beyond fixed vocabularies [40], language-conditioned localization provides scalable perception primitives [41], and region-centric image-language pretraining and prompting improve localization and transfer when regions are the natural unit of supervision [42, 43]. Coordinate scaffolding improves spatial alignment between language and vision [44], self-supervised visual backbones

improve invariance for robust grounding [45], and video-capable segmentation improves temporal consistency under motion and occlusion [46]. However, accurate segmentation alone does not yield stable tool contact: masks do not directly encode functional subregions, contact orientation, or lever-arm structure. RegioManip builds on region-centric perception, but makes regions interaction-ready by extracting contact primitives from grounded region geometry and lifting them into ranked SE(3) contact-frame hypotheses.

2.3. Safe Contact-Rich Manipulation

Contact-rich manipulation has long been studied through interaction control, force regulation, and stability-aware execution. Learning-based force control explicitly targets force and torque structure in robot policies [8, 9], while force-centric imitation learning models wrench signals from demonstrations [10]. Data scarcity remains a major challenge, and physics-driven data generation has been explored to synthesize broader contact interactions [47]. Visuotactile and contact-centric policy learning further show that tactile and force information become essential when vision alone is insufficient [11]. Safety-oriented approaches also emphasize stability mechanisms [48], passivity-based control to bound energy injection during interaction [49], and uncertainty-aware environment modeling for safer execution when failures are costly [14].

Analytical wrench reasoning provides another important foundation for safe tool interaction. Classical grasp-quality analysis characterizes admissible forces and torques using grasp wrench space and force-closure conditions [50, 51]. These methods provide principled, contact-geometry-dependent limits on the wrenches that a grasp or contact can sustain. RegioManip adopts this view by deriving its admissible force and torque budgets from the grasp wrench space of the resolved contact geometry, so that wrench limits follow from the inferred contact type rather than from fixed heuristics or manually tuned impedance alone.

Recent tool-manipulation systems increasingly combine perception, affordance reasoning, monitoring, and staged correction. ToolManip is a representative industrial system that combines modular VLM grounding, multimodal affordance representation, motion and force primitive reasoning, and execution monitoring for zero-shot industrial tool manipulation [4]. Other work explores task-adaptive representation selection [37], while process-oriented applications such as robotic screwdriving highlight the need for precise alignment and consistent interaction geometry under real hardware constraints [5, 52]. These systems demonstrate the importance of modular reasoning and execution feedback, but wrench outcomes are often handled through predefined motion/force primitive templates, execution monitoring, or reactive correction.

RegioManip aligns with the modular philosophy of recent tool-manipulation systems, but changes the executed representation. Instead of relying only on sparse keypoints, geometric constraints, predefined motion/force primitives, or reactive force correction, RegioManip derives admissible force and torque budgets from the grasp wrench space of the resolved contact, adapts constraint strictness according to grounding confidence, and triggers hierarchical repair from residual and wrench signals. This directly couples region-level grounding, contact-frame regulation, and enforceable wrench margins during sustained contact.

3. Methodology

3.1. Problem Formulation

The problem addressed in this work is zero-shot contact-rich tool manipulation. Given a single RGB-D observation and a natural-language instruction ℓ, the robot must determine how the tool should be grasped, where the resulting wrench should be applied, and how stable physical contact can be maintained throughout execution. No task demonstrations, object CAD models, or task-specific policies are assumed. Let $I \in \mathbb{R}^{H\times W\times 3}$ denote the RGB image, $D \in \mathbb{R}^{H\times W}$ the registered depth map, and $K = (f_x, f_y, c_x, c_y)$ the known camera intrinsics. A high-level plan [28] decomposes ℓ into a step sequence $S = \{\tau_i\}$, where each step τ_i specifies the executing entity, action mode, reference region, and a Boolean success predicate ϕ_i. As illustrated in Fig. 3, RegioManip maps (I, D, K, ℓ) to a physically grounded closed-loop manipulation policy through four coupled stages: region-level affordance grounding, geometry validation and lattice pruning, contact-frame hypothesis lifting, and wrench-bounded confidence-adaptive execution.

3.2. Region-Level Affordance Grounding

Given the observation $o_t = (I_t, D_t)$ and step context τ_i, this stage produces a control-facing region representation that is simultaneously semantic, geometric, and reliability-aware. The output is a region map $\mathcal{A}$ that supplies ranked affordance hypotheses for the downstream stages, as illustrated in Stage 1 of Fig. 3.

3.2.1. Object Localization and Segmentation

Given ℓ, Grounding DINO [53] localizes the tool and target objects as axis-aligned regions of interest using a detection confidence threshold $\tau_{\text{det}} = 0.35$. Within each region, SAM 2 [46] generates candidate binary masks $\{M_k\}$ in automatic mask generation mode, and masks smaller than 600 px^2 are discarded. The remaining candidate pairs (k, j) are merged only when

$$\text{IoU}(M_k, M_j) \geq 0.80 \;\wedge\; \text{MAD}(D_t \mid M_k \cap M_j) \leq \varepsilon_d \quad (1)$$

where $\text{MAD}(\cdot)$ is the median absolute deviation of depth within the overlap and $\varepsilon_d = 5$ mm. This depth-consistency condition prevents merging masks that overlap in the 2D projection but correspond to distinct 3D surfaces, a failure mode that appearance-based merging cannot detect.

3.2.2. Structured Affordance Representation

Rather than assigning an informal binary handle/work-end label as in prior keypoint-and-vector schemes [12, 13], we ground each candidate tool region into a structured

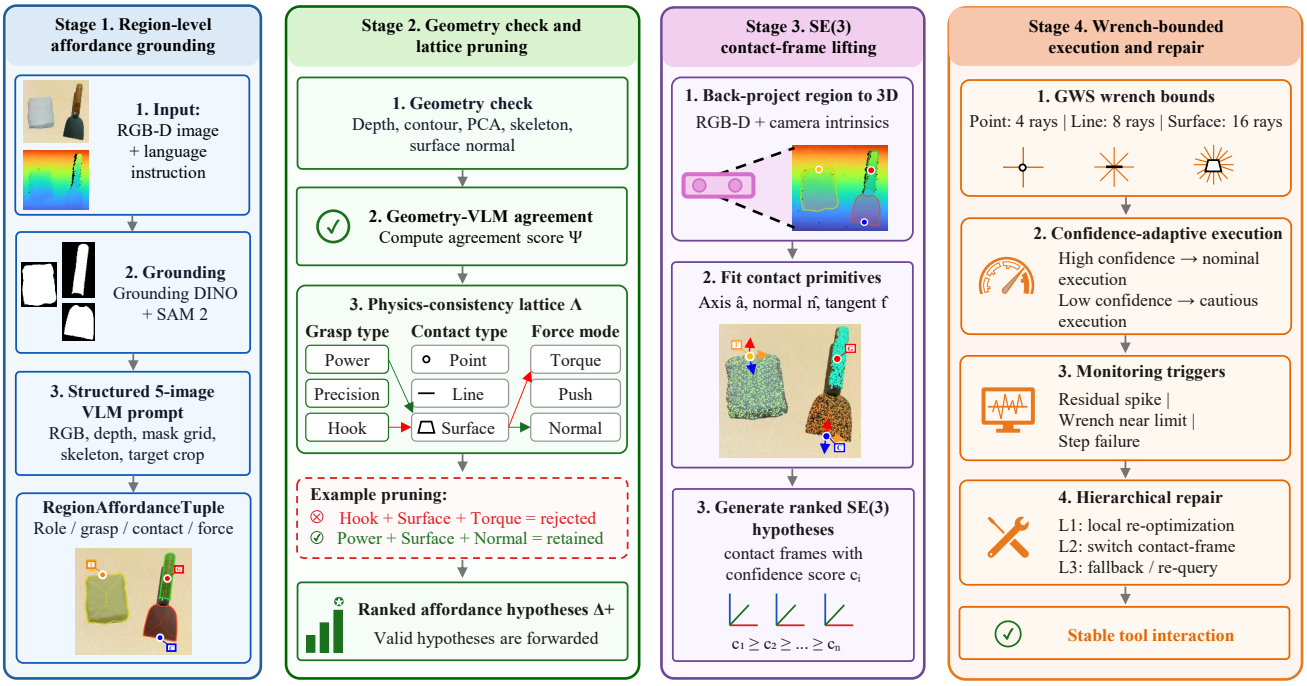


Figure 3: RegioManip pipeline. Stage 1 performs region-level affordance grounding from RGB-D input and language. Stage 2 applies geometry checking and physics-consistency lattice pruning. Stage 3 lifts valid regions to ranked SE(3) contact-frame hypotheses. Stage 4 executes with GWS-derived wrench bounds, confidence-adaptive control, and hierarchical repair.

RegionAffordanceTuple

$$a_p = \left(r_p,\ g_p,\ c_p,\ f_p,\ b_p,\ s_p,\ \rho_p,\ \{u_p^k\}\right) \tag{2}$$

where $r_p \in \{\text{handle}, \text{workend}\}$ is the functional role; $g_p \in \{\text{power}, \text{precision}, \text{pinch}, \text{hook}\}$ is the grasp type; $c_p \in \{\text{point}, \text{line}, \text{surface}\}$ is the contact geometry type; $f_p \in \{\text{push}, \text{pull}, \text{torque}, \text{normal}\}$ is the dominant force mode; $b_p \in \{\text{terminus}, \text{lateral}\}$ is the contact-face selector designating which work-end feature engages the target; s_p is a verb-conditioned semantic label such as $\text{scrape}_{\text{blade}}$; $\rho_p \in [0,1]$ is the aggregate confidence; and $u_p^k \in [0,1]$ is the calibrated per-attribute uncertainty for attribute $k \in \{g, c, f\}$. The contact geometry type c_p conditions the wrench synthesis stage, the contact-face selector b_p conditions contact-point extraction, and $\{u_p^k\}$ modulates execution strictness, making perceptual reliability an explicit input to control.

3.2.3. Structured 5-Image VLM Prompt

The full tuple a_p is predicted in a single GPT-4o [54] call using a structured five-image prompt, in which each view addresses a distinct perceptual gap: the RGB crop provides appearance context; the depth map supplies metric scale; the mask grid converts grounding into a structured selection problem; the skeleton overlay exposes tool elongation and contact-face geometry; and the target crop conditions c_p and f_p on the actual interaction surface. Mask geometry statistics are embedded as a structured table to calibrate the prediction against shape evidence, and predicting all attributes jointly in a single call ensures cross-attribute consistency while reducing the compounding errors that arise in sequential pipelines. From each valid mask the line primitive λ_p and patch primitive π_p are extracted and stored alongside a_p for downstream 3D lifting.

3.2.4. Geometry-VLM Agreement and Physics-Consistency Lattice

Concurrently with the VLM call, three geometry-only predictors are evaluated independently from mask statistics: grasp type $\hat{g}_{\text{geo}}$ from handle elongation and area, contact geometry $\hat{c}_{\text{geo}}$ from the work-end bounding-box aspect ratio, and force mode $\hat{f}_{\text{geo}}$ from a curated verb-mode lexicon applied to ℓ. Each prediction is lifted to a categorical distribution by scaling toward a one-hot vector in proportion to its confidence, and an analogous distribution is formed from the VLM prediction. The per-attribute agreement score is then defined as

$$\psi_k = 1 - \text{JSD}_2\left(P_k^{\text{geo}} \,\|\, P_k^{\text{vlm}}\right) \tag{3}$$

where JSD_2 is the base-2 Jensen-Shannon divergence [55], which is symmetric and bounded in $[0,1]$. This makes the agreement score sensitive to distributional disagreement rather than label mismatch alone.

Since not all combinations of (g_p, c_p, f_p) are physically realizable, mechanical constraints are encoded as a physics-consistency lattice

$$\Lambda : \Omega_g \times \Omega_c \times \Omega_f \to \{0, \varepsilon_\Lambda, 1\} \tag{4}$$

over the $4 \times 3 \times 4 = 48$ candidate triples. Forbidden triples ($\Lambda = 0$) include mechanically invalid combinations such as hook grasp with torque-dominant interaction, while improbable triples ($\Lambda = \varepsilon_\Lambda = 0.45$) include weakly

supported combinations such as pinch grasp with push-dominant interaction. We assume that the geometry predictor and VLM prediction are conditionally independent; therefore, their per-attribute distributions are fused by elementwise product, reweighted by Λ, and normalized to yield the ranked affordance simplex Δ^+.

The aggregate agreement score $\Psi = \text{mean}_k(\psi_k)$ and per-attribute uncertainty

$$u_p^k = 1 - \eta_k^{\text{vlm}} \cdot \psi_k \tag{5}$$

where η_k^{vlm} is the VLM confidence for attribute k, are propagated to the confidence-adaptive execution schedules in Stage 4. A score $\Psi < 0.40$, well below the observed mean agreement of 0.743, triggers a single-pass coherence-repair retry, targeting the low-agreement tail in which the geometry and VLM predictions substantively disagree. Committing to a single VLM prediction at grounding time, as in keypoint-based zero-shot methods [12, 13], discards this distribution and weakens downstream recovery. RegioManip instead forwards the full Δ^+ for physics-based multi-hypothesis selection. The output of this stage is the region map $\mathcal{A}$, where each record stores the mask, affordance tuple a_p, aggregate confidence ρ_p, validity indicator $v_p = 1$ iff $\rho_p \geq 0.45$, 2D primitives (λ_p, π_p), and geometric metadata. Only valid regions with $v_p = 1$ are forwarded to Stage 3.

3.3. Contact-Frame Hypothesis Lifting

Building on the region map $\mathcal{A}$, this stage lifts valid tool regions into ranked $\text{SE}(3)$ contact-frame hypotheses, as illustrated in Stage 3 of Fig. 3.

3.3.1. 3D Back-Projection and Primitive Fitting

For each canonical part with $v_p = 1$, mask pixels are back-projected to a 3D support set X_p using a pinhole model with an 8-px nearest-valid-depth search. The contact-face selector b_p designates the functional feature on the work-end mask, defining the contact point C^*; the grip point G is back-projected from the one-third-along-skeleton position on the handle mask. This selection is immune to off-axis appendages such as hammer claws that corrupt centroid-based estimates. From X_p, contact geometry is summarized via SVD-based PCA. The first singular vector defines the tool axis $\hat{a}$ and the third estimates the contact surface normal $\hat{n}$. The tangent is $\hat{t} = (\hat{a} \times \hat{n})/\|\hat{a} \times \hat{n}\|$, forming a right-handed orthonormal contact basis $W = (\hat{a}, \hat{n}, \hat{t})$. The metric grasp width w_g is estimated in 3D from the handle cloud as the binned 90th-percentile radial extent about $\hat{a}$, directly informing gripper aperture planning. Primitives with 90th-percentile fitting residual above 15 mm are excluded.

3.3.2. Temporal Fusion on Manifolds

Primitives estimated from individual frames fluctuate under depth noise and viewpoint variation. The grip point G_t and contact point C_t^* in $\mathbb{R}^3$ are smoothed by an exponential moving average with gain $\beta = 0.30$, reducing to raw per-frame values in single-frame deployment. The unit vectors $\hat{a}_t$ and $\hat{n}_t$ lie on S^2; we therefore apply spherical linear interpolation (SLERP) with the short-arc convention,

$$\text{Slerp}(\hat{u}, \hat{v};\, s) = \frac{\sin\big((1-s)\omega\big)}{\sin\omega}\,\hat{u} + \frac{\sin(s\omega)}{\sin\omega}\,\hat{v} \tag{6}$$

where $\omega = \arccos(\hat{u} \cdot \hat{v})$, $\hat{u}$ is the previous fused axis and $\hat{v}$ the current estimate ($s = 0.30$). SLERP moves along the great-circle arc at constant angular velocity and exactly preserves the unit-norm constraint, avoiding the directional bias of naive linear blending and preventing contact-frame drift under sustained interaction.

3.3.3. Contact-Frame Construction and Ranking

Candidate frames are constructed from the fused primitives, with the unsigned SVD eigenvectors resolved by enumerating sign hypotheses. Writing the closing and approach axes as $\varepsilon_t \hat{t}$ and $\varepsilon_n \hat{n}$ with $(\varepsilon_t, \varepsilon_n) \in \{\pm 1\}^2$, the 6-DoF grasp pose assembles the rotation $R = [\varepsilon_t \hat{t} \mid \hat{a} \mid \varepsilon_n \hat{n}]$, whose determinant is $\det R = \varepsilon_t \varepsilon_n$. Only the two combinations with $\varepsilon_t \varepsilon_n = +1$ are proper rotations; the remaining two are reflections and are discarded, guaranteeing $T_{\text{grasp}} \in \text{SE}(3)$. For a surface-directed approach along $-\hat{n}$ ($\varepsilon_n = -1$) this fixes the closing axis to $-\hat{t}$, giving $R = [-\hat{t} \mid \hat{a} \mid -\hat{n}]$; the retained rotation is converted to a quaternion via the Shepperd method [56]. Each surviving hypothesis is ranked by a composite confidence score

$$c_i = w_1 \Psi + w_2 \bar{\eta}_{\text{vlm}} + w_3\, p(h^*) + w_4\, q_{\text{geo}}, \quad \sum w = 1 \tag{7}$$

where $w_1 = w_2 = 0.30$, $w_3 = 0.25$, $w_4 = 0.15$; $\bar{\eta}_{\text{vlm}}$ is the mean VLM attribute confidence; $p(h^*)$ is the adjusted simplex probability; and $q_{\text{geo}} = \max(0,\, 1 - r_{P_{90}}/15\ \text{mm})$ is the primitive fitting quality. The two agreement-based terms, geometry–VLM consistency Ψ and mean VLM confidence $\bar{\eta}_{\text{vlm}}$, are weighted equally and highest as the primary reliability signals, while the hypothesis-probability and geometric-fitting terms receive lower weight as secondary evidence. Because the weights sum to one and each term lies in $[0, 1]$, $c_i \in [0, 1]$ directly, so the execution schedules of Stage 4 reach both their conservative and nominal endpoints. The ranked set $\mathcal{H}^+$, together with Δ^+, is forwarded to Stage 4.

3.4. Wrench-Bounded Confidence-Adaptive Execution

Given the ranked contact-frame hypotheses $\mathcal{H}^+$ and affordance simplex Δ^+, this stage compiles and executes a wrench-bounded action program with confidence-adaptive control and hierarchical repair, as shown in Stage 4 of Fig. 3.

3.4.1. Grasp Wrench Space Derivation

Grounded force and torque bounds are derived by constructing the Grasp Wrench Space (GWS) for the resolved contact geometry [50]. The friction coefficient $\mu = 0.40$ is a conservative value typical of dry metal-on-polymer tool contact, and the preload lower bound $f_{n,\min} = 0.25$ N reflects the minimum reliably commanded contact force on our hardware; both are lower bounds, so the derived wrench

budgets remain valid rather than optimistic. GWS extreme rays are constructed according to the contact type c_p: point contact yields 4 rays spanning the friction cone, line contact yields 8 rays with torsional moment pairs about the contact line, and surface contact yields 16 rays with moment pairs about both tangential axes. This contact-type-sensitive ray count directly couples c_p to the physically realizable wrench set, replacing the heuristic template scaling used in prior force-primitive methods [4]. Fig. 4 illustrates the corresponding contact abstraction across the grasp, align/contact, and execution stages.

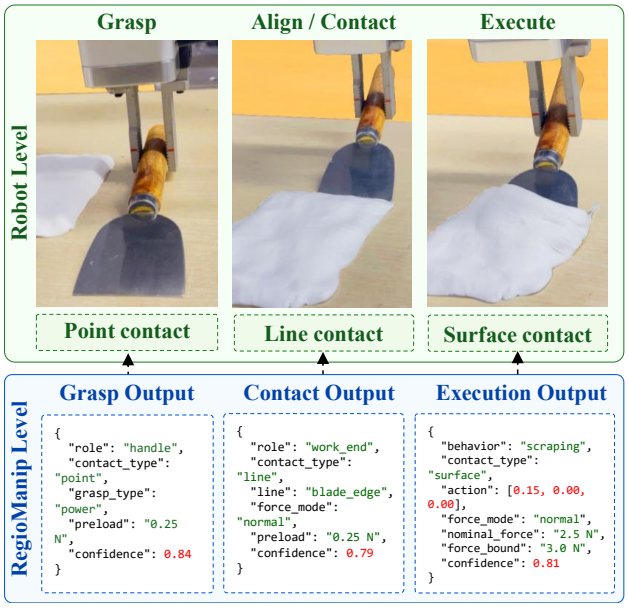


Figure 4: Contact abstraction across execution stages. Top: robot-level sequence from grasp to align/contact to execution. Bottom: corresponding RegioManip outputs, resolving the interaction as point, line, and surface contact with bounded force parameters.

Rather than committing to a single affordance triple at grounding time, each hypothesis in Δ^+ is evaluated in descending probability order. For each candidate, the GWS rays are recomputed using the corresponding contact geometry, and the task wrench demand w_{req} is checked for membership through a linear feasibility test. The required wrench w_{req} is formed in the contact frame from the selected force mode f^* and a nominal magnitude prior calibrated to the experimental setup, and is refined online from the wrist F/T reading during the contact-light approach. The selected hypothesis h^* is the highest-probability entry for which

$$w_{\text{req}} \in \text{conv}(\{w_k\}),$$

where $\{w_k\}$ denotes the GWS extreme rays.

3.4.2. Action Constraint Program

Given step τ_i and the selected hypothesis h^* with affordance triple (g^*, c^*, f^*), the mode-conditioned constraint program is compiled as

$$\Pi_i = \left(F_i,\ L_i,\ \mathcal{C}_i^{\text{term}},\ \mathcal{C}_i^{\text{path}},\ \mathcal{W}_i,\ \Theta_i\right) \tag{8}$$

where F_i contains the selected contact frame and top-K alternatives from $\mathcal{H}^+$; L_i stores the grounded action geometry from region primitives; $\mathcal{C}_i^{\text{term}}$ encodes terminal constraints such as alignment within ε_R; $\mathcal{C}_i^{\text{path}}$ encodes path constraints including monotonic progress along $\hat{t}$, wrench-limit compliance, and normal alignment; $\mathcal{W}_i$ is the GWS-derived admissible wrench set; and Θ_i contains confidence-conditioned scheduling parameters. The action mode m_i selects a residual template from four motion primitives: LinearMove for scraping, wiping, and sliding; RotateMove for screwing and loosening; and ApproachMove and RetractMove for safe contact-light approach and withdrawal.

3.4.3. Confidence-Adaptive Execution

The composite confidence score c_i modulates execution authority through two coupled schedules. The admissible wrench bound increases with confidence, while the geometric strictness weight increases under uncertainty:

$$\begin{aligned} w_{\max}(c_i) &= w_{\min} + c_i(w_{\text{nom}} - w_{\min}), \\ \lambda(c_i) &= \lambda_{\text{nom}} + (1 - c_i)(\lambda_{\text{safe}} - \lambda_{\text{nom}}) \end{aligned} \tag{9}$$

Because $c_i \in [0, 1]$, the wrench schedule spans from conservative to nominal, and the strictness schedule tightens as confidence decreases. Under low confidence, the system applies smaller admissible forces and stronger residual penalties, restricting motion to the most reliable directions. Under high confidence, it executes at nominal wrench and strictness settings. Caution thus becomes an explicit function of perceptual reliability rather than a fixed impedance parameter.

Execution proceeds in a receding-horizon loop. At each control cycle, the system updates c_i from the current monitoring state, refreshes the constraint program Π_i with the corresponding wrench bound and scheduling parameters, solves a short-horizon constrained optimization warm-started from the previous solution, applies the first segment, and replans at approximately 2.5 Hz. Recomputing c_i online allows the controller to tighten or relax execution authority as contact evidence accumulates during sustained interaction, rather than committing to a single confidence estimate fixed at grounding time.

3.4.4. Monitoring and Hierarchical Repair

Geometric convergence is quantified by an alignment-gap score $\bar{g}_i(t)$, which sums weighted positional and orientation mismatches between corresponding tool and target features. In parallel, the interaction wrench is sampled by the wrist-mounted F/T sensor at 100 Hz. The resulting wrench signal is summarized at each receding-horizon control cycle and compared against the admissible wrench set $\mathcal{W}_i$, while any high-rate violation beyond the hard wrench bound immediately triggers a protective stop.

When $\bar{g}_i(t)$ fails to decrease, residuals stagnate, or the wrench approaches its admissible bounds, RegioManip invokes an escalating three-level repair policy. The first level performs local re-optimization from the current warm start for up to three attempts. The second level switches to the

next-ranked contact-frame hypothesis in $\mathcal{H}^+$. The third level applies mode-level fallback through retreat-and-re-approach or coarse realignment.

A step is declared complete when its success predicate ϕ_i is satisfied: the terminal constraints are met and the residuals remain stably bounded over a 0.5 s dwell window, confirming that the intended contact relation has been maintained rather than only instantaneously reached. A protective stop is triggered if the measured wrench exceeds its bound or all repair levels are exhausted.

Algorithm 1 RegioManip closed-loop policy

Input: RGB-D observation $o = (I, D)$
Input: camera intrinsics K, instruction ℓ
Output: per-step pose $T \in \mathrm{SE}(3)$
Output: admissible wrench set $\mathcal{W}_i$
$S \leftarrow \mathrm{Plan}(\ell)$ ▷ step sequence $\tau_i = (e_i, m_i, o_i, \phi_i)$
for each step $\tau_i \in S$ **do**
  // Stages 1–2: region-level grounding + geometry check/lattice pruning
  $\{M_k\} \leftarrow \mathrm{SAM2}(\mathrm{GroundingDINO}(I, \ell))$
  merge masks ▷ Eq. (1)
  for each valid region p **do**
    $a_p \leftarrow \mathrm{VLM_5img}(I_{\mathrm{rgb}}, I_{\mathrm{depth}}, I_{\mathrm{grid}}, I_{\mathrm{skel}}, I_{\mathrm{tgt}})$ ▷ Eq. (2)
    $\Psi, \{u_p^k\} \leftarrow \mathrm{Agreement}(P^{\mathrm{geo}}, P^{\mathrm{vlm}})$ ▷ Eq. (3)
    $\Delta^+ \leftarrow \mathrm{Normalize}(\Lambda \odot P^{\mathrm{geo}} \odot P^{\mathrm{vlm}})$
  end for
  $\mathcal{A} \leftarrow \{p : v_p = 1\}$
  // Stage 3: contact-frame hypothesis lifting
  $X_p \leftarrow \mathrm{BackProject}(M_p, D, K)$
  $(\hat{a}, \hat{n}, \hat{t}), C^*, G, w_g \leftarrow \mathrm{FitPrimitives}(X_p)$
  fuse over frames ▷ EMA β, SLERP s
  $\mathcal{H}^+ \leftarrow \{\text{proper-rotation frames}\}$
  score c_i ▷ Eq. (7)
  // Stage 4: wrench-bounded confidence-adaptive execution
  build GWS rays for c_p
  $h^* \leftarrow \arg\max\, p(h)$ s.t. $w_{\mathrm{req}} \in \mathrm{conv}(\{w_k\})$
  compile Π_i ▷ Eq. (8)
  repeat ▷ receding horizon @ $\approx$ 2.5 Hz
    update c_i
    set $w_{\max}(c_i)$, $\lambda(c_i)$ ▷ Eq. (9)
    solve short-horizon program
    apply first segment
    monitor $\bar{g}_i$, wrench @ 100 Hz
    repair if needed
  until ϕ_i satisfied or safety stop
end for

4. Experiments and Results

4.1. Experimental Setup

RegioManip targets single-arm, contact-rich tool operations without CAD models, object meshes, or task demonstrations. We assume quasi-static scenes in which the tool and target remain stationary between observation and contact. The fixed eye-to-hand viewpoint is subject to partial self-occlusion by the arm during execution, which the monitoring and repair stages are designed to tolerate. Planning relies on cloud-based VLM inference and is performed offline, while closed-loop execution, monitoring, and repair run locally at the control rate.

Table 1
Key implementation hyperparameters.

Parameter	Value
Stage 1: Region-level affordance grounding	
DINO detection threshold τ_{det}	0.35
Minimum mask area	600 px^2
Mask-merge IoU threshold	0.80
Mask-merge depth threshold ε_d	5 mm
Region validity threshold ρ_p	0.45
Coherence-repair threshold Ψ	0.40
Improbable-triple weight ε_Λ	0.45
Stage 3: Contact-frame hypothesis lifting	
Depth search radius	8 px
Primitive residual threshold (P90)	15 mm
EMA gain β	0.30
SLERP parameter s	0.30
Confidence weights (w_1, w_2, w_3, w_4)	$(0.30, 0.30, 0.25, 0.15)$
Stage 4: Wrench-bounded confidence-adaptive execution	
Friction coefficient μ	0.40
Preload lower bound $f_{n,\min}$	0.25 N
GWS rays (pt / line / surf)	4 / 8 / 16
Re-optimization attempts	3
Success dwell window	0.5 s
Replanning rate	$\sim$2.5 Hz
F/T sampling rate	100 Hz

We validate RegioManip on a JAKA Zu3 6-DoF robotic arm equipped with a wrist-mounted force/torque sensor and an Intel RealSense D435i RGB-D camera. The benchmark contains six industrially motivated contact-rich tool-manipulation tasks: putty scraping, board cutting, roller cleaning, nail hammering, button pressing, and screw loosening. These tasks cover linear and rotational tool motions as well as constant-force and threshold-force interaction regimes. Each task is tested with two tool instances of different geometries, yielding twelve task-tool conditions. Each condition is evaluated over 20 independent trials, resulting in 240 trials per method. Initial tool and target poses are varied across trials within the reachable workspace. Together, the tasks cover the motion types and force regimes targeted by RegioManip while remaining small enough for dense repeated evaluation.

All methods receive only a single RGB-D observation and a natural-language instruction at inference time. In the open-world regime, no task demonstrations, object CAD models, task-specific training data, pose priors, or object meshes are provided; RegioManip is compared with MOKA and GeoManip. In the reference-assisted regime, a FoundationPose [16] pose-tracking prior is additionally provided, enabling comparison with ToolManip [4]. To avoid differences caused by VLM capability, all methods use GPT-4o when a VLM query is required.

We report four metrics: task success rate, rotation error, translation error, and peak normalized wrench utilization. A trial is successful when the task objective is completed, the intended contact relation is maintained during execution, and no protective stop is triggered. Rotation and translation errors measure deviations between the intended and executed contact frame during sustained contact. Peak normalized wrench utilization measures the maximum ratio between the measured interaction wrench and the GWS-derived admissible bound. For the three stability metrics, lower is better. Values are computed from representative samples during the active contact phase and averaged over the 20 trials per condition. For task success rates, 95% binomial confidence intervals are computed using Wilson score intervals over the 240 trials per method or variant.

Key implementation hyperparameters are summarized in Table 1. Hardware specifications, low-level controller parameters, force/torque sensor details, and camera calibration are provided in Appendix D. Evaluation uses a single manipulator, workspace, and camera; cross-platform and multi-camera validation remains future work, so the industrial claims reflect a representative workstation rather than guaranteed transfer across hardware.

4.2. Comparison with Baseline Methods

We compare RegioManip with two open-world baselines, MOKA [12] and GeoManip [7], and with ToolManip [4] in the reference-assisted regime. The overall task performance results are summarized in Tables 2 and 3, where bolded values indicate the best performance within each regime.

RegioManip achieves the highest average success rate and performs best on five of the six tasks, while matching GeoManip on screw loosening. This improvement comes from grounding functional regions into ranked $\mathrm{SE}(3)$ contact-frame hypotheses and enforcing GWS-derived wrench bounds during execution. As the ablation study reports in Table 5, removing confidence scheduling lowers success by 14.1 points, from 80.8% to 66.7%, and increases peak wrench utilization from 0.60 to 0.78. This indicates that confidence-adaptive wrench bounds contribute beyond the base pipeline by reducing excessive wrench usage under uncertain contact hypotheses.

Screw loosening remains the hardest open-world case because success depends on maintaining axial alignment while transmitting torque through a small contact region. In this task, small orientation errors can quickly move the required wrench close to the GWS boundary, leaving less margin for hypothesis switching and repair. This explains why RegioManip improves the average open-world performance but only matches GeoManip on screw loosening.

In the reference-assisted regime, RegioManip-FP achieves the highest average success rate, reaching 90.8% compared with ToolManip's 86.7%. This result shows that pose tracking improves execution, but does not remove the need for contact-frame-consistent wrench regulation. ToolManip benefits from pose anchoring and step-wise verification, whereas RegioManip-FP combines FoundationPose tracking with GWS-derived wrench bounds, confidence-adaptive strictness, and hierarchical repair. The 10.0-point gap between RegioManip and RegioManip-FP suggests that contact-frame uncertainty remains a major bottleneck in open-world manipulation, while the 4.1-point gap between RegioManip-FP and ToolManip suggests that wrench-bounded confidence-adaptive execution provides additional benefit beyond pose tracking alone.

Fig. 5 provides a qualitative comparison on the putty-scraping task, which requires sustained surface contact and consistent normal force. The top row shows the robot during active contact, and the bottom row shows the resulting surface condition inside the target region marked by the red dashed box. MOKA and GeoManip exhibit visible tool skew and incomplete surface coverage, consistent with their higher rotation errors and peak wrench utilization in Table 4. RegioManip maintains more stable blade alignment and produces cleaner surface coverage than both open-world baselines, while RegioManip-FP achieves the cleanest result by combining pose tracking with wrench-bounded, confidence-adaptive execution.

Table 4 further supports the stability benefit of RegioManip. In the open-world regime, MOKA produces the highest rotation error, 8.10°, and peak normalized wrench utilization, 0.85. GeoManip reduces rotation error to 7.10° but still shows high wrench utilization, 0.79. RegioManip lowers these values to 5.60° and 0.60 by coupling contact-frame lifting with GWS-derived bounds and confidence-adaptive strictness. In the reference-assisted regime, RegioManip-FP further reduces rotation error to 2.50° and translation error to 2.00 mm, improving on ToolManip's 3.00° and 3.00 mm. This indicates that explicit contact-frame regulation remains beneficial even when a strong pose prior is available.

Fig. 6 illustrates the confidence-adaptive behavior on the putty-scraping task. In both confidence regimes, the measured interaction wrench remains within the GWS-derived admissible bound [50]. Under high confidence, the controller permits nominal execution and produces a smoother force profile; under low confidence, it applies more conservative wrench bounds and stronger correction, resulting in cautious but slightly less smooth force evolution.

4.3. Ablation Study

To quantify the independent contribution of each component, we conduct a structured ablation across the same six-task benchmark under the open-world regime, with 240 trials per variant. Each variant removes exactly one component while holding the rest of the pipeline at its default configuration, so the resulting change in success and stability isolates that component's marginal contribution rather than reflecting interaction effects between multiple removed components. The open-world regime is used throughout because it exercises every stage of the pipeline without a pose prior, making the contribution of each perceptual and

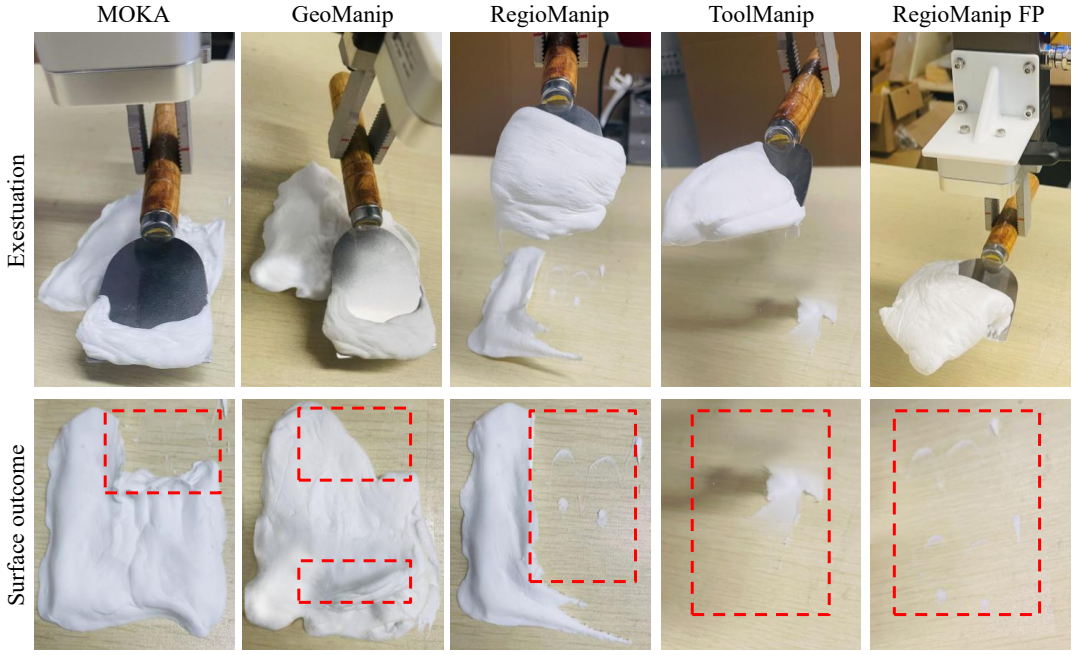


Figure 5: Qualitative execution comparison on the putty-scraping task. Top row: robot in active contact for each method. Bottom row: surface outcome within the target region (red dashed box). MOKA and GeoManip show tool skew and incomplete coverage, while RegioManip and RegioManip-FP maintain stable blade alignment and cleaner surface coverage. Surface outcomes are shown from different trial instances, so minor lighting variations may appear across methods.

Table 2
Task success rate, open-world regime (no pose prior, no CAD model). Bold indicates the best performance.

Force req.	Task	MOKA [12]	GeoManip [7]	**RegioManip (ours)**
Constant	Putty scraping	0.40	0.45	**0.75**
	Board cutting	0.50	0.60	**0.85**
	Roller cleaning	0.50	0.60	**0.90**
Threshold	Nail hammering	0.65	0.75	**0.80**
	Button pressing	0.75	0.80	**0.85**
	Screw loosening	0.65	**0.70**	**0.70**
Average	–	0.575 [0.512, 0.636]	0.650 [0.588, 0.708]	**0.808 [0.754, 0.853]**

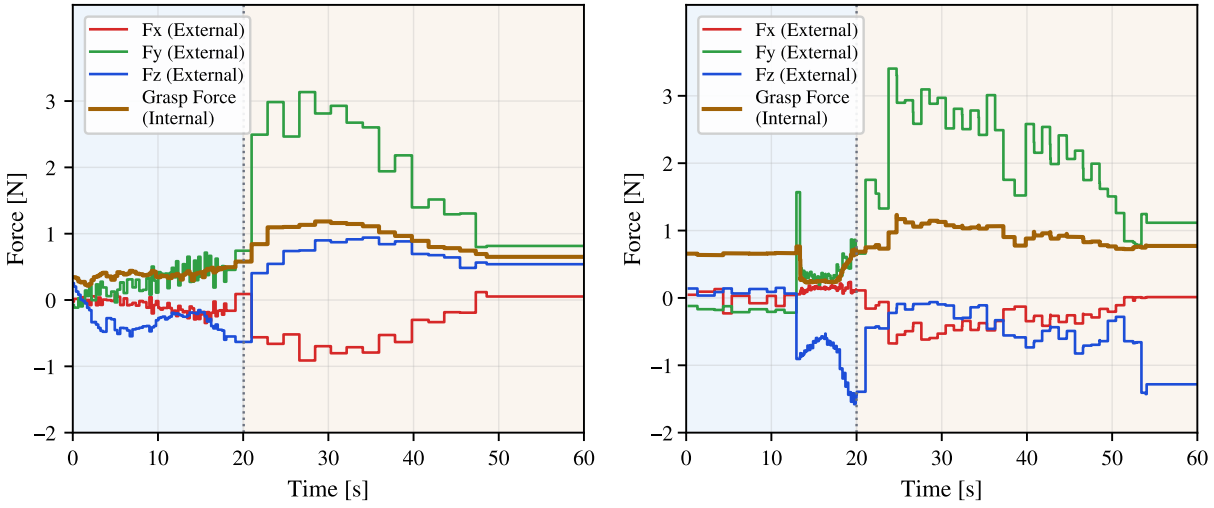
Table 3
Task success rate, reference-assisted regime (FoundationPose [16] pose prior enabled). Bold indicates the best performance.

Force req.	Task	ToolManip [4]	**RegioManip-FP (ours)**
Constant	Putty scraping	0.85	**0.90**
	Board cutting	**0.85**	**0.85**
	Roller cleaning	0.95	**1.00**
Threshold	Button pressing	**0.90**	**0.90**
	Nail hammering	0.85	**0.95**
	Screw loosening	0.80	**0.85**
Average	–	0.867 [0.818, 0.904]	**0.908 [0.865, 0.939]**

control component fully observable. Table 5 reports success and stability for all eight variants. Two patterns organize the results: components that gate or stabilize perception dominate task success, while confidence scheduling acts primarily on bounded wrench utilization.

Perception integrity is the load-bearing factor. Region validity gating is the single most important component; removing it collapses success to 38.3% and nearly doubles rotation error, with screw loosening falling to 2/20, because fragmented or depth-deficient regions corrupt every downstream stage in a way that neither confidence scheduling nor repair can compensate for. Structured guidance and the geometry-VLM agreement check protect the same grounding stage at lower cost (58.3% and 66.7%); both degrade most on scraping and screw loosening, the tasks with the smallest contact targets, while leaving hammering and pressing nearly intact. Complementary static-image evaluation confirms the agreement check detects genuine distributional divergence: with JSD enabled, mean $\Psi = 0.743$ against $\Psi = 1.0$ in the ablation, with coherence repair firing in 4.5% of trials at 1.5 s overhead.

Contact-frame stability depends on temporal fusion and the physics consistency lattice. Removing temporal fusion (60.0%) lets single-frame depth noise propagate into frame instability, degrading the sustained-contact tasks most severely (scraping 11/20, screw loosening 7/20). Removing the 48-triple lattice (63.3%) admits mechanically forbidden



(a) High confidence (b) Low confidence

Figure 6: Measured external contact forces (F_x, F_y, F_z) and internal grasp force during the putty-scraping execution shown qualitatively in Fig. 5, under (a) high-confidence and (b) low-confidence scheduling. The shaded region marks the sustained-contact phase. In both cases the internal grasp force remains stable and external forces stay within the GWS-derived admissible bound, but the low-confidence schedule shows a wider, more oscillatory excursion band, reflecting its more conservative, repair-prone execution.

combinations such as hook × torque and torque × surface, raising peak wrench utilization to 0.74 and producing excursions on screw loosening (7/20 vs. 14/20) and hammering (13/20 vs. 16/20). The repair policy is the dominant recovery mechanism; without it success falls to 48.3%, most sharply on roller cleaning (11/20 vs. 18/20) and screw loosening (4/20 vs. 14/20), where contact drift accumulates gradually and becomes unrecoverable without re-optimization or hypothesis switching. Fig. 7 illustrates the three repair levels in action: under high confidence no repair is needed; under medium confidence a single re-alignment step suffices; under low confidence both local re-optimization and hypothesis switching are invoked before stable contact is achieved.

Confidence scheduling behaves differently from the rest. It leaves success comparatively high (66.7%) yet is the largest single driver of wrench utilization, raising peak wrench utilization from 0.60 to 0.78. The controller exhibits higher wrench utilization even on completed trials when confidence conditioning is removed, confirming that the schedule of Eq. (9) supplies a protective margin that is largely independent of task success.

4.4. Robustness, Limitations, and Runtime

We further examine RegioManip under practical disturbances that commonly arise in real workspaces, focusing on perceptual robustness, representative failure modes, and runtime feasibility.

Perceptual robustness. RegioManip is evaluated under three observation conditions (normal view, changed viewpoint, and clutter); success rates are reported as mean task-level percentages across the six-task benchmark. Quantitative results are summarized in Table 6, and a qualitative grounding comparison is shown in Fig. 8. This robustness test evaluates perceptual grounding under visual disturbance rather than full end-to-end task execution; therefore, Table 6 is not directly comparable to the task-execution success rates in Tables 2 and 3. RegioManip outperforms all three baselines across all three conditions. The performance gap is most pronounced under clutter, where RegioManip achieves 91.7% compared to 50.0% for ReKep, 60.0% for MOKA, and 30.0% for GeoManip. ReKep and MOKA predict sparse keypoints and relational anchors from appearance alone, which become unreliable when the viewpoint changes or distractors occlude the tool. GeoManip infers geometric axes and part labels, performing well in clean views (86.0%) but collapsing under clutter (30.0%) because its geometric cues are sensitive to unfiltered background structure. RegioManip instead grounds each region into a structured mask with depth-consistency gating and lift-readiness validation before committing any hypothesis, making the final grounding substantially more stable across all three observation conditions, as visible in Fig. 8.

Failure analysis. Failures decompose into four categories. Ambiguous region grounding in cluttered scenes accounts for 38% of failures, most commonly on tools where the functional end is visually more prominent than the grip end, causing the VLM and geometry predictors to disagree on role assignment despite a high aggregate confidence. Contact-frame estimation failures account for 29%, arising primarily on tools with thin or highly reflective surfaces where depth noise corrupts 3D primitive fitting and produces degenerate or high-residual contact frames that are rejected by the validity gate. Execution failures not resolved by the hierarchical repair policy account for 24%, concentrated on screw loosening under high torque demand where wrench requirements consistently approach the GWS boundary. The remaining 9% involve partial occlusion and instruction ambiguity that cannot be recovered at the grounding stage.

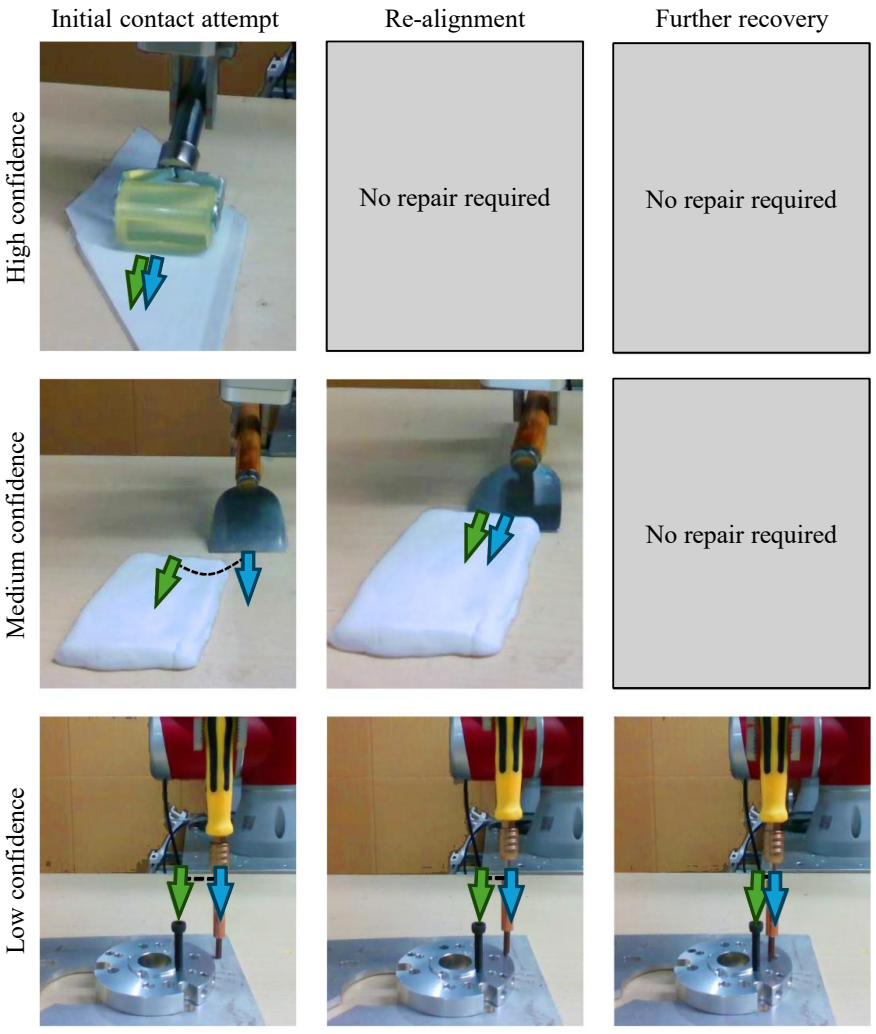


Figure 7: Hierarchical repair behavior under three confidence levels. High confidence requires no repair; medium confidence triggers a single re-alignment step; low confidence invokes local re-optimization followed by hypothesis switching before stable contact is achieved.

Table 4
Execution stability metrics across both regimes. Lower is better for all metrics; rotation error in degrees, translation error in mm.

Method	Regime	Rot. err. (°) ↓	Trans. err. (mm) ↓	Peak wrench util. ↓
MOKA [12]	Open-world	8.10	11.00	0.85
GeoManip [7]	Open-world	7.10	8.00	0.79
RegioManip (ours)	Open-world	**5.60**	**4.50**	**0.60**
ToolManip [4]	Reference-assisted	3.00	3.00	0.66
RegioManip-FP (ours)	Reference-assisted	**2.50**	**2.00**	**0.58**

Table 5
Ablation study: task success rate and execution stability. Bold indicates the best performance; lower is better for stability metrics.

Variant	Success (%)	95% CI (%)	Rotation error (°)	Translation error (mm)	Peak wrench util.
RegioManip (Full)	**80.8**	[75.4, 85.3]	**5.60**	**4.50**	**0.60**
w/o Geometry–VLM agreement	66.7	[60.5, 72.3]	7.10	5.90	0.71
w/o Physics consistency lattice	63.3	[57.1, 69.2]	7.60	6.30	0.74
w/o Confidence scheduling	66.7	[60.5, 72.3]	7.40	5.80	0.78
w/o Temporal fusion	60.0	[53.7, 66.0]	7.80	6.50	0.69
w/o Structured guidance	58.3	[52.0, 64.4]	8.20	7.10	0.72
w/o Repair policy	48.3	[42.1, 54.6]	9.60	8.20	0.84
w/o Region validity gating	38.3	[32.4, 44.6]	11.20	9.40	0.88

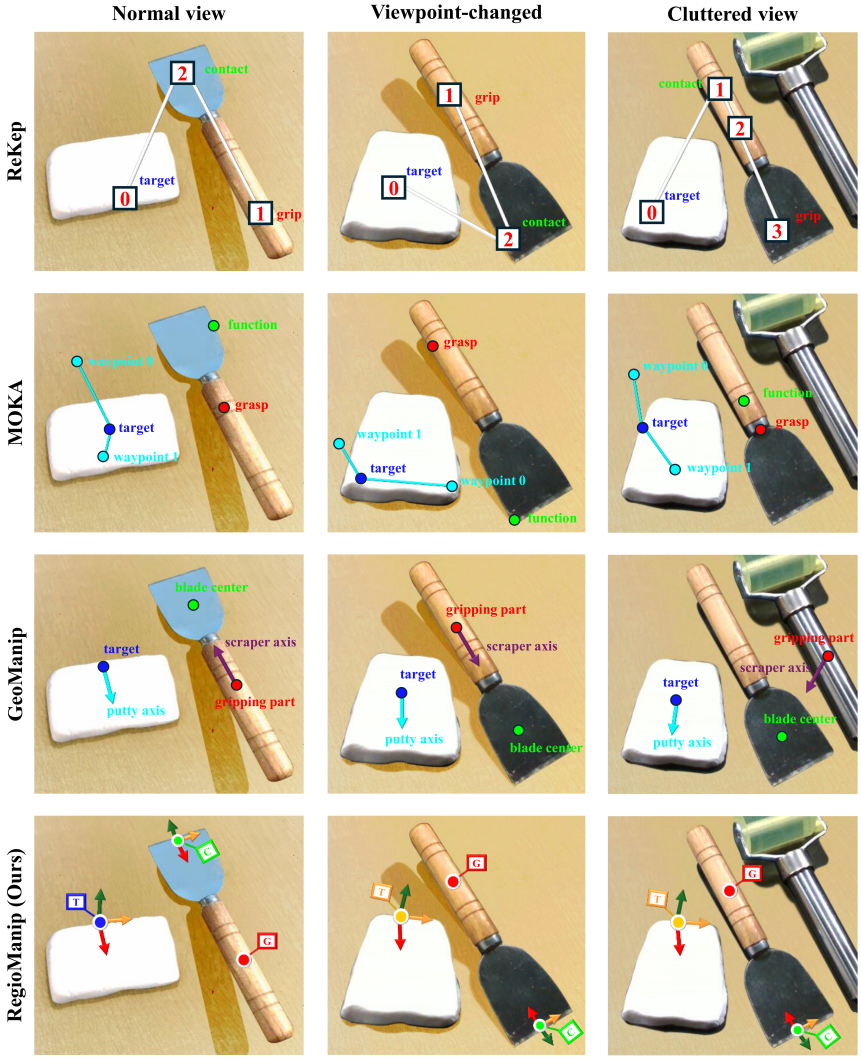


Figure 8: Qualitative grounding comparison under three observation conditions (normal view, viewpoint-changed, cluttered). RegioManip outputs region masks with grip (G, red) and contact (C, blue) points, while ReKep, MOKA, and GeoManip degrade in region identity and spatial consistency under viewpoint change and clutter.

Table 6
Perceptual robustness: mean task-level success rate (%) under three observation conditions across the six-task benchmark.

Method	Normal view	Viewpoint-changed	Cluttered
ReKep [13]	66.0	56.0	50.0
MOKA [12]	86.0	70.0	60.0
GeoManip [7]	86.0	76.0	30.0
RegioManip (ours)	**98.3**	**95.5**	**91.7**

Novel tool generalization. Despite these failure cases, the pipeline generalizes successfully to a broad set of novel tool instances not present in the benchmark. Fig. 9 shows eight representative tools spanning substantially different shapes and morphologies, including elongated, flat, brush-like, bent, and multi-part structures. Across all instances, the method correctly assigns grip (G, red) and contact (C, blue) points from the structured region representation, without any task-specific training or object model. These results suggest that the region-first affordance representation generalizes beyond the six-task benchmark to a broader range of tested tool geometries.

Runtime. Table 7 reports mean wall-clock time per manipulation step, measured over 50 scraping trials on the full system. The dominant cost is GPT-4o structured prediction (18.2 s), consistent with published multi-image VLM inference latencies. The geometry-VLM agreement check and coherence repair together add 1.5 s, less than 7% of total offline planning time, confirming that the uncertainty quantification mechanism carries negligible computational overhead. Baselines were not instrumented for wall-clock timing under an identical protocol; the dominant cost, the 18.2 s VLM call, is incurred by the VLM-based baselines as well, so RegioManip's added stages (under 1.5 s) are a small fraction of planning time rather than a categorical overhead. All remaining pipeline stages complete in under 1 s combined. Per-cycle execution at approximately 0.4 s supports

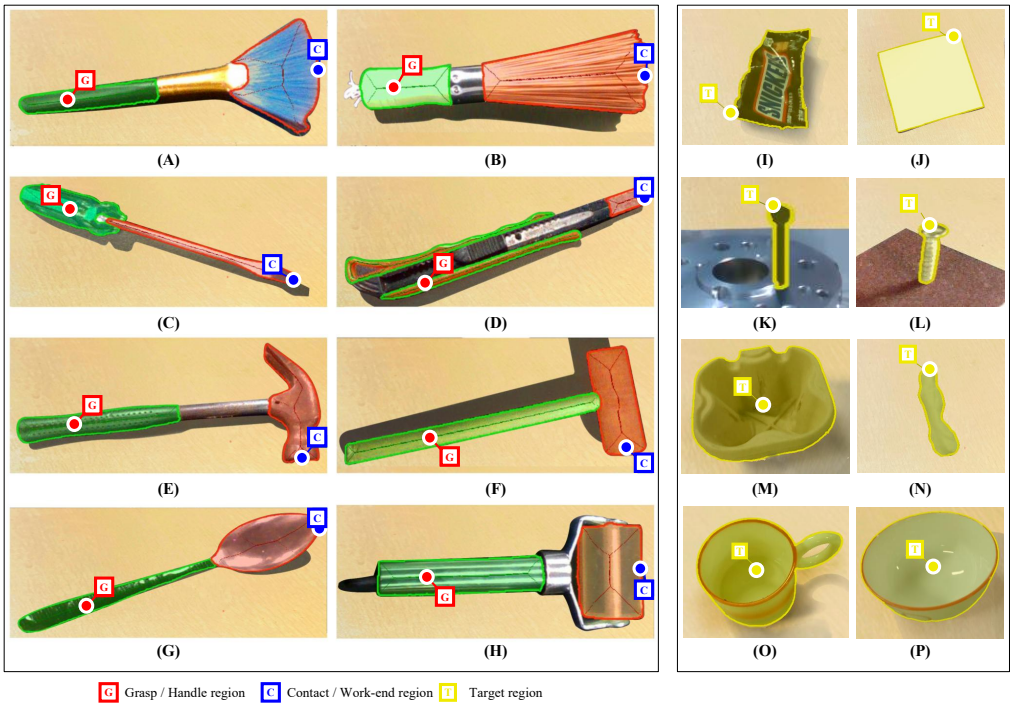


Figure 9: Novel tool generalization on eight unseen tool instances outside the six-task benchmark: (A) duster, (B) paint brush, (C) screwdriver, (D) box cutter, (E) claw hammer, (F) mallet, (G) spoon, (H) roller. The pipeline correctly identifies grip (G, red) and contact (C, blue) points in each case. Target-region examples are marked as (T, yellow): (I) trash, (J) paper, (K) bolt, (L) screw, (M) tray, (N) putty, (O) cup, and (P) bowl.

2.5 Hz receding-horizon control, which is sufficient for pre-plannable industrial tasks such as surface processing and precision assembly in fixed workstations.

Table 7: Mean per-step runtime breakdown (50 scraping trials).

Stage	Mean time (s)
Object detection (Grounding DINO [53])	0.8
Mask generation (SAM 2 [46])	1.4
VLM structured prediction (GPT-4o [54])	18.2
Geometry-VLM agreement + coherence repair	1.5
3D contact-frame lifting	0.6
Constraint program compilation	0.3
Per-cycle receding-horizon optimization	0.4
Total (offline planning)	∼23
Per-cycle execution update	∼0.4

Two system-level limitations warrant explicit acknowledgment. First, offline planning depends on cloud API inference for GPT-4o; local deployment of a compact vision-language model would substantially reduce planning latency without requiring any change to the methodology.

Second, the system is sensitive to depth quality on specular and translucent surfaces, which produces unreliable 3D primitive fitting and accounts for 29% of observed failures. Future work will explore active illumination strategies, multi-return depth sensors, and multi-view or shape-prior-guided segmentation to improve coverage on reflective tool geometries.

5. Conclusion

We presented RegioManip, a zero-shot framework that converts grounded tool regions into wrench-bounded action constraints for contact-rich manipulation. Instead of outputting sparse keypoints or waypoints, RegioManip grounds each task-relevant region into structured affordance attributes in a single VLM call, cross-checks them against geometry-only estimates, and prunes mechanically infeasible combinations through a physics-consistency lattice. This produces a ranked distribution of affordance hypotheses rather than a single commitment. Valid regions are then lifted into ranked SE(3) contact-frame hypotheses that preserve orientation and lever-arm structure, while execution enforces force and torque budgets derived from the grasp wrench space of the resolved contact under confidence-adaptive strictness and residual- and wrench-triggered hierarchical repair.

Across six industrial tool operations on a JAKA Zu3, RegioManip improved task success and interaction stability over recent training-free baselines in both open-world and reference-assisted regimes. The results show that RegioManip reduces geometric drift and peak wrench utilization rather than compensating for misalignment through excessive force. Ablation studies further indicate that perception gating and frame stabilization dominate task success, while confidence scheduling primarily governs peak wrench utilization and adds negligible runtime overhead.

The main limitations are the latency of cloud-based VLM inference and sensitivity to depth quality on specular

and translucent surfaces. Future work will pursue compact local vision-language models to reduce planning latency, together with active illumination and multi-view or shape-prior-guided segmentation to improve coverage on reflective and non-convex tool geometries. These extensions may support broader deployment of wrench-bounded, reliability-aware tool manipulation in industrial settings.

CRediT authorship contribution statement

Dourjoy Debnath Rup: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Shakir Muhammad Afzal ul Haq:** Software, Data curation. **Wenzhe Zheng:** Software, Data curation. **Lijin Yang:** Software, Data curation. **Sen Du:** Software, Visualization. **Dhali Md Sabuj:** Writing – review & editing. **Qimin Li:** Conceptualization, Methodology, Supervision, Resources, Funding acquisition, Writing – review & editing. **Fuqiang Liu:** Resources, Data curation, Validation, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex for language editing, grammar checking, and manuscript consistency review. The authors reviewed and edited the content as needed and take full responsibility for the publication.

Appendix

A. Structured Five-Image VLM Prompt Template

This appendix gives the structured five-image GPT-4o prompt described in Section 3.2.3, which predicts the full *RegionAffordanceTuple* $a_p = (r_p, g_p, c_p, f_p, b_p, s_p, \rho_p, \{u_p^k\})$ (Eq. (2)) in a single call. Table 8 gives the prompt text.

Table 8
Prompt template for region-level affordance grounding.

Given the attached RGB crop of the candidate region, the registered depth map, a mask grid enumerating candidate sub-region masks indexed 0, 1, 2, …, a skeleton overlay exposing tool elongation and contact-face geometry, a crop of the target object or surface, and a table of mask geometry statistics (area, aspect ratio, elongation), assign to region $\{N\}$ a `role` (one of handle, workend), a `grasp_type` (one of power, precision, pinch, hook), a `contact_geometry` (one of point, line, surface), a `force_mode` (one of push, pull, torque, normal), a `contact_face` (one of terminus, lateral, indicating which work-end feature engages the target), and a verb-conditioned `semantic_label` (e.g. scrape_blade) for task instruction {instruction} and object category {object_category}. Predict all attributes jointly so that role, grasp, contact geometry, and force mode remain mutually consistent, and report an aggregate `confidence` and a per-attribute `attribute_confidence` for `grasp_type`, `contact_geometry`, and `force_mode`, each in $[0, 1]$. The required output format is: ⟨role⟩, ⟨grasp_type⟩, ⟨contact_geometry⟩, ⟨force_mode⟩, ⟨contact_face⟩, ⟨semantic_label⟩, ⟨confidence⟩, ⟨attribute_confidence⟩.

The predicted `grasp_type`, `contact_geometry`, and `force_mode` fields are each compared against the corresponding geometry-only estimate $\hat{g}_{\text{geo}}, \hat{c}_{\text{geo}}, \hat{f}_{\text{geo}}$ via the categorical JSD agreement score of Eq. (3) (Section 3.2.4), and the returned per-attribute confidence values supply η_k^{vlm} in that computation.

B. Per-Task Ablation Sensitivity

Note: This appendix is supplementary diagnostic evidence from an earlier experimental protocol and is *not* part of the main evaluation. Table 5 in the main text reports success and stability aggregated across the six-task benchmark for all eight ablation variants under the 240-trial protocol that constitutes the main evaluation. This appendix adds per-task breakdown for the five variants common to an earlier 120-trial, single-tool-instance protocol that preceded the introduction of the geometry-VLM agreement check and the physics-consistency lattice; it is included solely to illustrate *which* tasks each component protects. Values here are not directly comparable to Table 5: they come from a different (earlier, smaller) trial set. The two newest components (geometry-VLM agreement, physics-consistency lattice) are absent from this earlier protocol and so are not shown per-task; their aggregate contribution appears only in Table 5.

Removing temporal fusion degrades scraping and screw loosening most severely (11/20 and 7/20), where precise contact-frame orientation accumulates over sustained motion. Removing structured guidance most affects scraping

Table 9
Per-task trials succeeded out of 20 (earlier 120-trial protocol, five variants).

Variant	Scraping	Cutting	Rolling	Hammering	Pressing	Screw
Full (this protocol)	17/20	17/20	18/20	16/20	14/20	10/20
w/o Temporal fusion	11/20	13/20	15/20	12/20	14/20	7/20
w/o Structured guidance	10/20	12/20	14/20	14/20	14/20	6/20
w/o Confidence scheduling	10/20	14/20	14/20	14/20	16/20	6/20
w/o Repair policy	9/20	14/20	11/20	10/20	10/20	4/20
w/o Region validity gating	8/20	10/20	12/20	8/20	6/20	2/20

and screw loosening (10/20 and 6/20), tasks requiring precise linear-stroke or axial alignment, while hammering and pressing are comparatively unaffected. Removing the repair policy degrades rolling and screw loosening most (11/20 and 4/20), where contact drift accumulates gradually and becomes unrecoverable without re-optimization. Removing region validity gating produces the broadest degradation, with screw loosening dropping to 2/20, confirming that perceptual noise from weak upstream regions propagates through every downstream stage.

C. Worked Action Constraint Program Examples

This appendix gives two fully instantiated examples of the action constraint program $\Pi_i = (F_i, L_i, \mathcal{C}_i^{\text{term}}, \mathcal{C}_i^{\text{path}}, \mathcal{W}_i, \Theta_i)$ (Eq. (8)): putty scraping, a constant-force linear-contact task, and nail hammering, a threshold-force rotational task. Numeric values are illustrative of the compiled structure, consistent with the GWS construction of Section 3.4.1 and the confidence schedule of Eq. (9). In the verbatim listings below, `Pi_i` denotes Π_i and `W_i` denotes $\mathcal{W}_i$, matching the notation of Eq. (8).

C.1. Example 1: Putty Scraping

```
=== Pi_i (Putty Scraping, contact stroke) ===
F_i  (Contact frame):
  origin : back-projected centroid of blade edge
           t_i = [0.312, -0.047, 0.184] m (camera frame)
  x-axis : stroke direction e_x = [0.997, 0.012, -0.073]
  y-axis : surface normal e_y = [0.072, -0.003, 0.997]
  z-axis : e_z = e_x x e_y = [-0.014, 0.999, 0.002]

L_i  (Grounded action geometry):
  mode          : linear_contact
  stroke_vector : 0.08 m along e_x
  contact_depth : 0.003 m into surface normal e_y
  approach_dist : 0.05 m pre-contact standoff along e_y

C_i^term (Terminal constraints):
  1. blade-center within 3 mm of stroke end point
  2. surface alignment angle < 4 deg
  3. residual contact force in e_y in [4.0, 12.0] N

C_i^path (Path constraints):
  1. lateral deviation along e_z < 4 mm throughout stroke
  2. contact normal force F_y in [3.0, 15.0] N
  3. joint angle limits satisfied at every waypoint

W_i  (Admissible wrench set, confidence c_i = 0.87):
  w_max(c_i) = w_min + c_i (w_nom - w_min)
  Force bounds  : F_x, F_y <= 13.4 N,  F_z <= 20.4 N
  Torque bounds : T_x, T_y <= 1.04 Nm, T_z <= 1.04 Nm

Theta_i (Confidence-conditioned execution parameters):
  c_i                  : 0.87
  confidence_level     : high (nominal execution)
  repair_level_1       : local re-optimization, warm start
  repair_level_2       : switch to next-ranked hypothesis
  repair_level_3       : retreat-and-reapproach fallback
  success_predicate    : terminal constraints met AND
                         residuals bounded over 0.5 s dwell
```

C.2. Example 2: Nail Hammering

```
=== Pi_i (Nail Hammering, strike phase) ===
F_i  (Contact frame):
  origin : centroid of nail-head contact point
           t_i = [0.278, 0.031, 0.173] m (camera frame)
  x-axis : strike direction (downward) e_x = [0.008, -0.005, -0.999]
  y-axis : lateral stability axis e_y = [0.994, 0.107, 0.007]
  z-axis : e_z = e_x x e_y = [-0.108, 0.994, -0.006]

L_i  (Grounded action geometry):
  mode          : rotational_strike
  strike_vector : 0.04 m rapid deceleration along e_x
  approach_dist : 0.06 m pre-contact standoff along -e_x
  dwell_time    : 0.05 s post-contact dwell at peak force

C_i^term (Terminal constraints):
  1. striker face within 2 mm of nail-head centroid
  2. strike alignment: hammer-face normal within 5 deg of nail axis
  3. peak contact impulse over dwell exceeds threshold

C_i^path (Path constraints):
  1. lateral deviation < 3 mm during approach
  2. joint velocity limits not exceeded during strike
  3. wrist torque below safety limit throughout motion

W_i  (Admissible wrench set, confidence c_i = 0.74):
  w_max(c_i) = w_min + c_i (w_nom - w_min)
  Force bounds  : F_x, F_y <= 19.3 N, F_z <= 31.5 N
  Torque bounds : T_x, T_y <= 1.48 Nm

Theta_i (Confidence-conditioned execution parameters):
  c_i                  : 0.74
  confidence_level     : medium (moderate correction)
  repair_level_1       : re-align approach angle, warm start
  repair_level_2       : switch to alternative nail-head hypothesis
  repair_level_3       : retreat, re-acquire target, re-approach
  success_predicate    : terminal constraints met AND
                         no bounce residual > 3 mm after dwell
```

D. Hardware and Controller Details

Low-level controller. Cartesian impedance control with online replanning at the ∼2.5 Hz receding-horizon rate of Section 4.1 (Table 1); position gains $K_p = 800$ N/m (translation), $K_r = 50$ Nm/rad (rotation), critically damped with $D = 2\sqrt{K_p}$. The contact-normal axis is regulated by a PI force controller updated at 100 Hz from the F/T sensor, matching the sensor sampling rate in Table 1. A safety stop triggers when $|F| > 50$ N or $|T| > 5$ Nm on any axis. Inverse kinematics use a Jacobian pseudo-inverse solver with null-space regularization, resolved at each replanning step.

Camera extrinsics and calibration. The RealSense D435i is mounted eye-to-hand at a fixed position overlooking the workspace. The camera-to-robot-base transform $T_{\text{cam}\to\text{base}}$

is calibrated prior to all experiments using a 7×7 checkerboard with 25 mm squares via the standard OpenCV hand-eye calibration procedure, verified by reprojecting known end-effector poses into the camera frame and checking pixel-level alignment. Factory intrinsics are $f_x = 614.9$, $f_y = 615.0$, $c_x = 424.5$, $c_y = 240.1$ (pixels, at 848×480); depth is aligned to the color frame using the onboard hardware alignment module.

Table 10: Hardware specifications for all real-robot experiments.

Component	Specification
Robot arm	JAKA Zu3, 6-DoF, 590 mm reach, 3 kg payload
Force/torque sensor	JK-SE-II-200, $\pm$200 N / $\pm$20 Nm, 0.05 N resolution
Gripper	HEPG40-100 two-finger parallel, 40 mm stroke
RGB-D camera	Intel RealSense D435i, 848$\times$480 @ 30 Hz, FOV 87°$\times$58°
Workstation	Intel i7-12700F, 16 GB RAM, NVIDIA RTX 3080 GPU
Communication	JAKA SDK over Ethernet, 100 Hz joint command rate

E. Physics-Consistency Lattice

The lattice Λ assigns each of the $4 \times 3 \times 4 = 48$ affordance triples (g, c, f) one of three values: allowed (1), improbable ($\varepsilon_\Lambda = 0.45$), or forbidden (0). The assignments follow five mechanical rules. First, a hook grasp cannot grip to transmit a moment, so hook with torque is forbidden. Second, a hook supports axial pulling rather than pushing or pressing, so hook with push or normal is improbable while hook with pull is allowed. Third, a pinch grasp cannot reliably transmit torque regardless of contact type (improbable); however, surface contact enables stable push and normal force (allowed), line contact enables normal but not push, and point contact limits both push and normal to improbable. Fourth, a precision grasp is fully admissible across all contact and force modes, including torque on point contact as required by screw-type tasks. Fifth, a power grasp is admissible across every contact and force mode. Table 11 lists the complete lattice.

Table 11: Physics-consistency lattice Λ over all $4 \times 3 \times 4 = 48$ affordance triples. Entries are allowed (1), improbable ($\varepsilon_\Lambda = 0.45$), or forbidden (0).

Grasp	Contact	Push	Pull	Torque	Normal
Power	Point	1	1	1	1
	Line	1	1	1	1
	Surface	1	1	1	1
Precision	Point	1	1	1	1
	Line	1	1	1	1
	Surface	1	1	1	1
Pinch	Point	0.45	1	0.45	0.45
	Line	0.45	1	0.45	1
	Surface	1	1	0.45	1
Hook	Point	0.45	1	0	0.45
	Line	0.45	1	0	0.45
	Surface	0.45	1	0	0.45

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